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RESEARCH

AlphaNiftyAI : conquering the Indian stock market with AI

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Abstract

AlphaNiftyAI is a large language model (LLM) driven financial intelligence system designed to analyze Indian equities and generate actionable investment signals, and gives justification for each signal. It integrates four different sources of financial data that are price trends, company fundamentals, news sentiment, and macroeconomic context into a structured, explainable decision-making framework. AlphaNiftyAI employs a multi-agent pipeline architecture where each agent performs domain-specific summarization using the Llama3-70B model served via GROQ's high-speed inference API. One of the key innovations in AlphaNiftyAI is the incorporation of historical news and its impact, facilitated by FAISS, which enables the retrieval of past events similar to current news headlines. These past events are related to price changes that have happened, allowing the model to ground its sentiment analysis in historical performance data. Additionally, chain-of-thought prompting is adopted in the signal generation agent to produce clear and logically justified BUY/SELL/HOLD decisions, each accompanied by an LLM-generated confidence score on a scale of 1 to 10. The framework supports both weekly and monthly decisions, generating signals for all NIFTY 50 stocks. These signals are then backtested using multiple strategies, which include market-cap-based allocation, stop loss integration, and confidence score-based selections. AlphaNiftyAI assists users by providing AI-analyzed trading signals with risk consideration strategies, enabling them to make more informed and explainable investment decisions and achieve superior risk-adjusted returns. The best-performing strategy achieved returns of 69.23% over a 15-month period and 127.80% over a 2-year period, significantly outperforming the NIFTY benchmark during both periods.

Keywords AlphaNifty AI · Large language models · Indian stock market · Investment signals · GROQ · Llama3 · FAISS · Chain-of-thought prompting · Financial NLP · Backtesting

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1 Introduction

Financial markets are constantly evolving and are influenced by a combination of quantitative data, news sentiment, and macroeconomic indicators. Traditional quantitative models often rely on numerical data and static rules, which limit their ability to capture important information hidden in unstructured sources, such as financial news. Recent advancements in large language models (LLMs) have opened new possibilities in stock market analysis, enabling systems that can understand, reason, and act on textual information in a human-like way (Yang et al., 2023).

AlphaNiftyAI brings the capabilities of generative AI into financial research for the Indian stock market. The foundation of this work drew inspiration from recent research (Fatouros et al., 2024), which demonstrated the use of LLMs to generate monthly stock trading signals. AlphaNiftyAI initially reproduced this base approach for NIFTY 50[1] companies using monthly summaries of price, fundamentals, news, and macroeconomic data to generate `BUY/SELL/HOLD` signals with a chain-of-thought prompting technique. It was observed that monthly signal generation often missed key price peaks, where profit could be gained, and market volatility windows. To address this, the work was made into a more granular, weekly signal generation pipeline capable of giving more timely and accurate signals.

AlphaNiftyAI is designed as a modular, agent-based system, with each agent focused on a specific dimension of financial data:

- *Price Agent:* Computes weekly return, volatility, Sharpe ratio, drawdown, and peer comparison.
- *Fundamental Agent:* Summarizes valuation and profitability metrics such as P/E, return on equity (ROE), and profit growth from the latest available filings.
- *News Agent:* Uses a FAISS (Meta AI, 2025)-based similarity search to retrieve past news articles semantically close to current headlines, linking them to their historical next-week price impact.
- *Macro Agent:* Adds contextual awareness using the latest macroeconomic trends and indicators.

The outputs of all agents are passed to a Signal Agent, which uses the Llama3.3-70B (Meta AI, 2024) model served via the GROQ API[2] and a structured chain-of-thought prompt to generate `BUY/SELL/HOLD` signals, along with a model-derived confidence score from 1 to 10. AlphaNiftyAI supports both weekly operations and stores natural-language summaries and signals for each NIFTY 50 company. These signals are tested using various portfolio strategies. The system was backtested over several months of NIFTY 50 data. The best-performing strategy achieved a great return, significantly outperforming the NIFTY index benchmark.

AlphaNiftyAI demonstrated the power of large language models not just as summarizers, but as reasoning agents that can provide explainable, data-driven investment insights and trustworthy decisions. Its shift from monthly to weekly signals,

[1] https://www.nseindia.com/

[2] https://groq.com/

including the historical impact of news, and the addition of confidence-based and market-cap-based strategies, make it a unique contribution to the field of AI-driven financial systems, especially within the Indian market context. The contribution of our work can be summarised as follows:

- Developed an agent-based architecture (Price, Fundamental, News, and Macro Agents) to cleanly structure and summarize different financial data.
- Added a FAISS-based news retrieval system to find historically similar news articles with known next week impact on price, increasing the quality of the sentiment signal.
- Using a Signal Agent that uses Llama3-70B as an LLM with chain-of-thought prompting technique to generate reasonable BUY/SELL/HOLD decisions along with a confidence score (1–10).
- Identified problem with monthly signal generation frequency (missed price peaks) and changed to a more granular weekly signal generation pipeline.
- Designed and tested strategies like:
  - *NIFTY++*: A stoploss integrated strategy on the BUY signals by the LLM.
  - *MS-Top10/5-Cap-LLM-NIFTY++*: Strategies based on LLM-generated confidence scores and market cap in addition to NIFTY++.

The rest of the paper is structured as follows: Sect. 2 reviews the related works, highlighting existing approaches in financial forecasting and the role of large language models in that field. Section 3 presents the proposed methodology, which includes data collection, agent description, and signal generation strategies. Section 4 presents the results and discussion, where we evaluate the performance of Alpha NIFTY AI against benchmark indices and various mutual funds and analyze its effectiveness. Finally, the paper concludes with key insights.

2 Related work

The intersection of natural language processing (NLP) and financial forecasting has become a major focus with the rise of large language models (LLMs). Earlier, financial models focused on structured numerical data such as historical prices, ratios, and macroeconomic indicators. However, the growing ability of LLMs to process and reason over unstructured text, including news articles, earnings call transcripts, and policy statements, has opened new opportunities for signal generation and improving portfolio decisions.

An important study in this area is *Can Large Language Models Beat Wall Street?* by Fatouros et al. (2024), which presents a method that uses market data, news, and company information to make stock predictions. Their results indicate that this approach outperforms standard models on key metrics, including the Sharpe ratio and overall returns. BloombergGPT (Wu et al., 2023) is another major development in the field. It is trained on 700 billion tokens of financial and general text, resulting in significant improvements across various financial NLP tasks. Although it is not publicly available, its results highlight the value of domain-focused training. Similarly, FinBERT (Araci, 2019) adapts BERT for finance-specific sentiment tasks, marking an early step toward the LLM pipelines used today.

Financial forecasting has increasingly added artificial intelligence and deep learning models, with recent studies showing improvements over traditional statistical methods. For example, Zhang et al. (2024) shows that LLM-enhanced sentiment extraction significantly improves short-term return predictability compared to lexicon-based sentiment measures, especially when the market is unstable. This aligns with earlier findings that textual information has additional predictive power beyond price data alone.

News and media signals remain central to understanding the dynamics of asset prices. Aouadi et al. (2023) documents that real-time news sentiment has a significant impact on both volatility and returns, with stronger reactions observed for close attention and large-cap stocks. Similarly, Jiang et al. (2021) demonstrates that transformer-based language models outperform classical methods in capturing the complex relationships between news and price movements, underscoring the importance of modern NLP architectures in financial prediction.

Price-based signals also continue to receive strong empirical validation. The influential study by Fischer and Krauss (2018) demonstrates that deep neural networks trained solely on historical price data outperform traditional machine learning baselines across major global indices. More recent work by Tsantekidis et al. (2020) confirms that LSTM and CNN architectures effectively capture short-horizon dependencies in returns that linear models fail to exploit, supporting the integration of short-term price-trend information in AI-driven trading systems.

Fundamental indicators provide additional predictive values for medium and long horizons. Hou et al. (2021) provides strong global evidence showing that profitability, investment patterns, and quality metrics predict future returns across markets. Their findings emphasize the continued relevance of features such as ROE, leverage, and revenue growth signals commonly used in hybrid AI strategies.

Macroeconomic factors also influence stock market performance. Chen et al. (2022) shows that inflation surprises, monetary policy announcements, and global risk sentiment significantly change return dynamics, especially during regime shifts. This suggests that adding macroeconomic data to predictive pipelines enhances robustness under changing market conditions.

Finally, combining multiple data types has been shown to improve portfolio performance. Krauss and Do (2023) find that multimodal deep-learning architectures integrating price, textual, and fundamental data outperform single-source models and are more resilient during volatile periods.

In contrast to earlier approaches, some studies focus on prompt design and explainability. Liu et al. (2024) introduced LLM-Trader, which uses a chain-of-thought (CoT) prompting to produce both decisions and the reasoning behind them. Following the same idea, our system generates both decision and LLM confidence scores using CoT prompting, enabling better risk management.

Lastly, backtesting and benchmarking have become increasingly important in this rapidly developing field. Studies like GPTStockPicker (Zhou et al., 2023) and ReLLM (Zhang et al., 2023) demonstrate the need for thorough testing using historical stock returns and realistic trading conditions, rather than relying on fixed sentiment datasets. Following this, we developed a modular backtesting system that measures excess returns, drawdowns, Sharpe, and Sortino ratios across various strategies.

In summary, our work builds upon earlier research by integrating advancements in large language models, financial time series analysis, and macroeconomic modeling into a unified system. We introduce AlphaNiftyAI, a fully automated and modular pipeline that utilizes advanced large language models (LLMs), such as Llama-3 70B, along with company-specific datasets including company fundamentals, stock prices, financial news, and macroeconomic indicators. A key feature in our approach is a past price-impact memory module, which enhances news summarization by identifying similar historical news events and their corresponding market responses. Additionally, AlphaNiftyAI generates signals at a weekly level, enabling more detailed and precise backtesting across multiple strategies.

3 Methodology

AlphaNiftyAI implements a fully automated multi-agent investment pipeline that collects weekly stock prices, company news, fundamental data, and macroeconomic indicators. Each data type is processed by a summarization agent, which includes Price, News, Fundamental, and Macroeconomic agents, to extract key information, compute relevant metrics, and create their summaries. These summaries are generated by the signal agent, which utilizes a large language model with a chain-of-thought prompting technique to produce BUY/SELL/HOLD decisions, along with a confidence score ranging from 1 to 10. The confidence score indicates the degree of confidence the LLM has in its decision. The generated signals are backtested using next week's price data under multiple strategies, and performance is evaluated against the NSEI benchmark and other Nifty50 mutual funds using annualized volatility, Sharpe ratio, Sortino ratio, maximum drawdown, and win rate. The overall methodology is illustrated in the Fig. 1.

3.1 Data collection

The AlphaNiftyAI system ingests data from multiple financial domains, including price movements, fundamentals, news, and macroeconomics, each of which is essential for generating accurate and explainable investment signals. To maintain consistency, all data is collected on a weekly basis, organized into a standardized folder hierarchy by week and ticker symbol. All the data, except the historical news, is from the period June 2023 to May 2025. This section describes the specific datasets and their preprocessing for each agent.

3.1.1 Price data for price agent

Historical stock price data is widely used in financial analysis. Patterns in returns, volatility, and momentum extracted from this data can be used for predicting the value of future price movements. Recent studies show that machine learning models

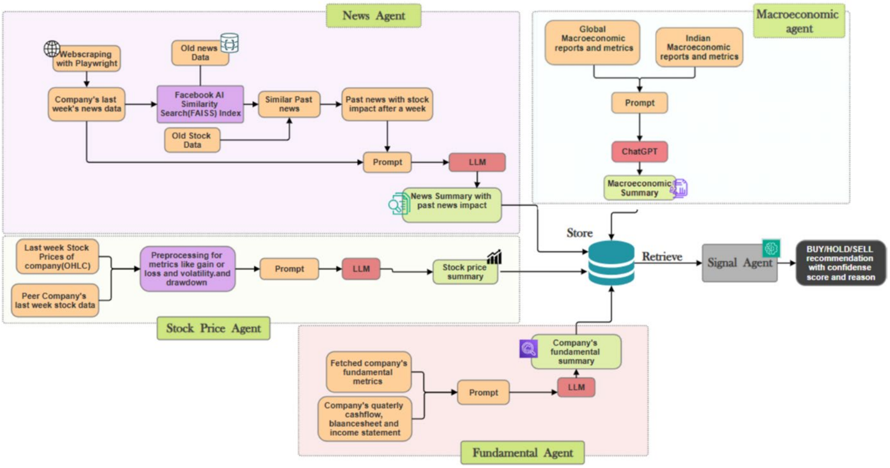

Fig. 1 Methodology

trained on price histories can capture market changes and improve short-term forecasting accuracy (Zhang et al., 2020). Therefore, price data remains a reliable foundation for quantitative modeling, and it should be considered an important factor.

The weekly price data forms the backbone of the Price Agent's computations. For each NIFTY 50 company, daily OHLCV (Open, High, Low, Close, Volume) data is collected for the five trading days of the week. This data is fetched using the `yfinance`[3] Python library. Historical stock prices have been collected using the same library.

3.1.2 Fundamental data for fundamental agent

Company fundamentals, including earnings, revenue growth, leverage, and profitability, play a crucial role in shaping investor expectations about a company's future market performance. Stocks with strong and improving fundamentals tend to attract higher expected returns, while those with weak fundamentals tend to experience decreasing valuations. Recent studies show that accounting-based signals (e.g., profitability, investment, and growth quality) consistently predict stock returns across global markets (Hou et al., 2020).

Fundamental data is fetched from the `yfinance` Python library, which provides updated metrics such as the price-to-earnings (P/E) ratio, Return on equity (ROE), return on capital employed (ROCE), debt-to-equity (D/E) ratio, net profit, sales growth, and EBITDA. Since fundamental metrics are typically updated monthly or quarterly, the most recent snapshot is aligned with the week being analyzed. Each company's fundamental data is saved as a JSON file. This allows the Fundamental Agent to extract and summarize the financial health and valuation of each company during signal generation.

[3] https://pypi.org/project/yfinance/

3.1.3 News data for news agent

Financial markets react quickly to new information, and news often becomes the first signal that shifts market direction. Positive developments, such as strong earnings, tend to boost investor confidence in investing in the company, while negative news triggers caution or risk-off behaviour, thus lessening the investor's tendency to invest in the company. Even when fundamentals stay unchanged, news can influence buying or selling pressure almost instantly. Prior research also shows that news sentiment has a measurable impact on stock returns and volatility, making its important role in shaping short-term market dynamics (Tetlock, 2007).

We implemented a browser automation script designed to collect the most recent news headlines, as traditional web scraping techniques were insufficient. Many modern news websites now use client-side rendering techniques and rate limits to prevent data extraction, making it difficult to scrape content directly. To overcome this challenge, we utilized Playwright, a browser automation tool that interacts with web pages in the same way a human would with a browser, effectively bypassing rendering restrictions. Our Playwright-based script uses Google News to gather the most recent news headlines on each company, which are continuously updated from the current date. This method yielded a dataset of approximately 12,000 news articles from various websites, covering all NIFTY 50 companies, from June 2023 to May 2025.

3.1.4 Historical news data collection

We collected corporate announcements and financial news data[4] from reputable platforms including *financialexpress.com*, *news18.com*, *businesstoday.in*, *moneycontrol.com*, *zeenews.india.com*, and *economictimes.com*. The dataset was structured according to our project requirements and includes the following fields:

- *Title:* The headline of the news article.
- *URL:* The web address of the article.
- *Description:* A concise overview of the article content.
- *Time:* The timestamp including date and time of publication.
- *Stocks Affected:* The companies mentioned in the article.

A total of 160,000 news articles of NIFTY 50 companies were initially collected. To ensure high data quality and relevance, we manually removed articles that were generic or irrelevant. After correction, the final dataset consists of 32,366 curated news articles from the period January 2021 to December 2023.

3.1.5 Macroeconomic data collection

Company fundamentals such as profitability, leverage, and growth indicators play a crucial role in explaining long-term stock performance because they reflect the com-

[4] https://desiquant.streamlit.app/

pany's true economic strength. Recent incidents demonstrate that financial ratios, such as book-to-market, earnings quality, and investment patterns, possess significant predictive power for future returns (Novy-Marx, 2013; Hou et al., 2015). These data help investors differentiate between undervalued and overvalued companies, making fundamentals an important input for quantitative models.

AlphaNiftyAI utilizes macroeconomic data from official sources, including the Reserve Bank of India (RBI), the World Bank, and the National Stock Exchange (NSE). The collected macro data informs the model about broader economic trends and their significance, particularly in cases where company performance is influenced by external macroeconomic factors.

The following indicators were collected monthly, quarterly, or annually, depending on their availability and publication schedule:

- *Foreign Exchange Reserves:* Sourced from the RBI's official data portal,[5] giving insights into the country's financial stability and trade capacity.
- *GDP Growth, Unemployment, and Inflation:* Obtained from the World Bank's data repository[6] offering long term economic health indicators.
- *FII/DII Investments:* Collected from NSE reports[7] capturing investor sentiment and capital flows into Indian equities.
- *Currency Exchange Rate:* Reference exchange rates were sourced from the RBI's daily rate archive.[8]
- *Annual Reports and Economic Surveys:* Additional indicators such as detailed inflation breakdowns,[9] GDP composition,[10] and Foreign Direct Investment (FDI) flows[11] were extracted from various annual publications by the RBI.[12]

3.2 News summary agent

The News Summary Agent (Fig. 2) is responsible for analyzing recent company-specific news and evaluating its likely market impact, incorporating both the content of the news and historical reactions to similar headlines. This agent combines real-time news collection, semantic similarity search, and language model reasoning to provide context-aware summaries.

To begin, the latest news headlines for each NIFTY 50 company are scraped weekly using a browser automation script implemented with Playwright. Once recent headlines are collected, the agent searches for semantically similar news from the past using a FAISS-based vector similarity index. The FAISS index was constructed using a curated historical news corpus spanning 2021 to 2023, comprising over

[5] https://www.rbi.org.in/

[6] https://data.worldbank.org/indicator

[7] https://www.nseindia.com/reports/fii-dii

[8] https://www.rbi.org.in/Scripts/ReferenceRateArchive.aspx

[9] https://www.rbi.org.in/Scripts/AnnualReportPublications.aspx?Id=1449

[10] https://www.rbi.org.in/Scripts/AnnualReportPublications.aspx?Id=1447

[11] https://www.rbi.org.in/Scripts/AnnualReportPublications.aspx?Id=1454.A

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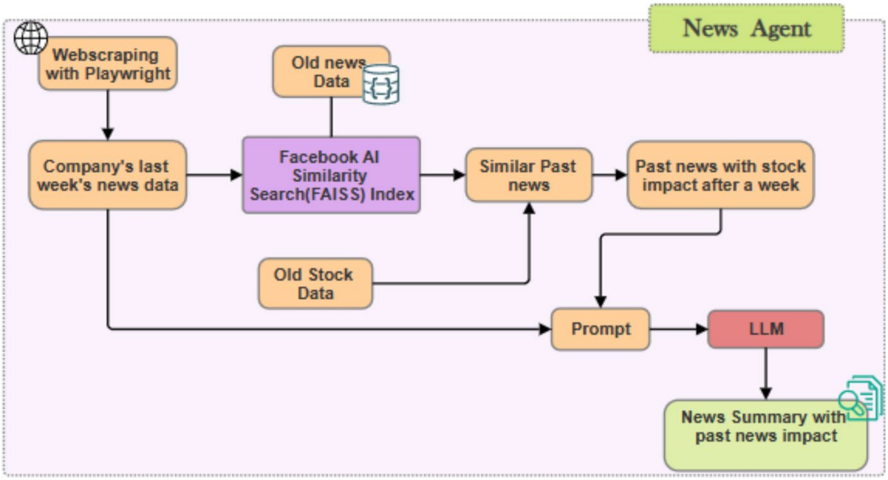

Fig. 2 News agent architecture

Table 1 Example of news summary

News summary for Adani Enterprises Ltd
News sentiment: Negative Notable Events:
* Regulatory risks may impact hospital stocks, including Medanta, KIMS, Apollo, Max, and Narayana, according to analyst views.
* Health risks for hospitals amid SC diktat and steep valuations may also affect hospital stocks.
Tone: The news tone is negative for the stock, suggesting potential downward pressure on hospital stocks due to regulatory risks and health risks. The historically similar news also indicates a negative impact on stock prices, with an average impact of around −3.5%.

32,000 manually filtered financial news headlines. Each past headline is paired with the actual stock return in the following week, allowing us to quantify its historical price impact. The similarity search module retrieves the top 3 closest historical headlines for each news article. Along with each similar article, we extract the observed price movement of the relevant stock during the week after news publication. This enables us to develop a proxy understanding of how similar news has previously impacted the market. This collection of recent news and retrieved historical context is compiled into a structured prompt, which is passed to the Llama3.3-70B language model hosted via GROQ's API. The prompt explicitly separates sections such as recent news articles for the current week, Similar past news headlines, and the associated next week price impact of those past headlines. The LLM processes this prompt and generates a concise, natural language summary of the company's news sentiment as shown in Table 1 for the week. This output is stored in the text file path and later used by the Signal Agent in the final decision-making step.

3.2.1 FAISS (Facebook AI similarity search)

FAISS, as illustrated in Algorithm-1, is a powerful library for fast similarity search in large-scale high-dimensional data. It is widely used for tasks like semantic search, clustering, and recommendation systems, where finding similar items quickly is very important. In AlphaNiftyAI, FAISS is used to search for similar chunks of text in the large dataset of financial news articles. Since text embeddings are high-dimensional, brute-force search is computationally expensive. FAISS helps by storing chunk embeddings in an index named IndexFlatL2, which enables fast retrieval based on Euclidean distance. Whenever a query is entered, it is encoded into an embedding, and then it finds the most similar chunks. These chunks are then mapped back to their original articles, allowing for to retrieval of relevant news efficiently and quickly.

Input: News dataset $\mathcal{D}$, Pre-trained Embedding Model $\mathcal{M}$, User query q
Output: Top k relevant news articles

Step 1: Indexing Phase
Initialize FAISS index $\mathcal{I}$;
Initialize chunk-to-article mapping $\mathcal{A}$;
for all article $a \in \mathcal{D}$ **do**
  Split a into chunks $\{c_1, c_2, ..., c_n\}$;
  for all chunk $c \in a$ **do**
    $e \leftarrow \mathcal{M}.\texttt{encode}(c)$ ▷ Generate embedding
    $\mathcal{I}.\texttt{add}(e)$ ▷ Store in FAISS
    $\mathcal{A}[e] \leftarrow a$ ▷ Map chunk to article
  end for
end for

Step 2: Search Phase
$q_e \leftarrow \mathcal{M}.\texttt{encode}(q)$ ▷ Encode query
$(\mathcal{D}_q, \mathcal{I}_q) \leftarrow \mathcal{I}.\texttt{search}(q_e, k')$ ▷ Retrieve k' similar chunks
Initialize article relevance scores $\mathcal{S}$;
for all $(idx, score) \in (\mathcal{I}_q, \mathcal{D}_q)$ **do**
  $a \leftarrow \mathcal{A}[idx]$ ▷ Find corresponding article
  if $a \notin \mathcal{S}$ **then**
    Initialize $\mathcal{S}[a] = (0, -\infty, 0)$ ▷ (total score, min score, chunk count)
  end if
  Update $\mathcal{S}[a]$:
   $\mathcal{S}[a].\texttt{total} \mathrel{+}= score$
   $\mathcal{S}[a].\texttt{min} = \min(\mathcal{S}[a].\texttt{min}, score)$
   $\mathcal{S}[a].\texttt{count} \mathrel{+}= 1$
end for

Step 3: Ranking and Output
Sort articles in $\mathcal{S}$ by $(\mathcal{S}[a].\texttt{min}, -\mathcal{S}[a].\texttt{count})$
Return top k articles with their matched chunks

Algorithm 1 FAISS-Based News Retrieval

3.3 Stock price agent

The Stock Price Agent (Fig. 3) analyzes short-term market behavior based on weekly OHLC (Open, High, Low, Close) stock price data. It extracts technical signals such as returns, volatility, drawdowns, and relative performance, which are later summarized by an LLM to inform the final trading signal.

Weekly stock data for each NIFTY 50 company is stored, covering Monday to Friday prices. Peer company data is also used to compute sector-relative metrics.

The following key indicators are computed:

- *Weekly Return (R):* Measures the overall price change across the week.

$$R = \frac{P_{\text{close}} - P_{\text{open}}}{P_{\text{open}}} \times 100 \tag{1}$$

where:

- P_{open}: Stock price on Monday (opening of the week)
- P_{close}: Stock price on Friday (closing of the week)
- R: Return percentage for the week

- *Volatility* (σ): It represents the variation of daily returns within the week.

$$\sigma = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(r_i - \bar{r})^2} \tag{2}$$

where:

- r_i: Return on day i
- $\bar{r}$: Average daily return in the week
- n: Number of trading days (typically 5)
- σ: Standard deviation of returns (volatility)

- *Maximum Drawdown (MDD):* Measures the largest single drop from a peak to a trough during the week.

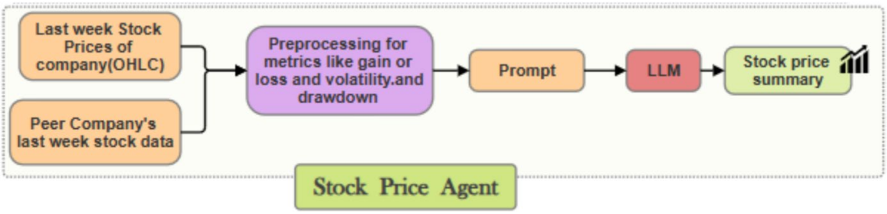

Fig. 3 Stock price agent architecture

$$\text{MDD} = \max_{t\in[1,n]} \left(\frac{\max_{s\in[1,t]} P_s - P_t}{\max_{s\in[1,t]} P_s} \right) \times 100 \tag{3}$$

where:

- P_t: Price at time step t
- $\max_{s\in[1,t]} P_s$: Highest price observed up to time t
- MDD: Maximum percentage drop within the week

- *Peer Average Return* (R_{peer}): Measures how a company's return compares to the average return of its sector peers.

$$R_{\text{peer}} = \frac{1}{m} \sum_{j=1}^{m} R_j \tag{4}$$

where:

- R_j: Weekly return of peer company j
- m: Number of peers in the same sector
- R_{peer}: Average peer return for comparison

- *Sharpe Ratio* (S): Indicates how well the stock compensated for risk (Sharpe, 1966).

$$S = \frac{R - R_f}{\sigma} \tag{5}$$

where:

- R: Weekly return of the stock
- R_f: Risk-free rate (assumed to be 0 for weekly analysis)
- σ: Volatility(standard deviation of returns)
- S: Risk-adjusted return (Sharpe Ratio)

All computed metrics are made into a structured prompt that explains the stock's weekly behavior. The prompt includes peer comparisons and asks the LLM to summarize bullish/bearish trends, volatility shifts, and notable movements. The LLM-generated summary (as shown in Table 2) is saved, and this serves as an input to the signal agent for final prediction.

3.3.1 Peer company identification

To understand how the company's stock performs in context, we examined the returns of peer companies. Identifying the right peers reveals whether a price change is unique to a company or impacts the entire sector. For each NIFTY 50 company,

Table 2 Stock price summary

Stock price summary for Adani Enterprises Ltd.
ADANIENT's weekly performance was not good, with a return of −0.27% and a Sharpe ratio of −0.03, indicating poor risk-adjusted returns. The stock's volatility was moderate at 32.04%, which is relatively high compared to its peers. In comparison to its peers, ADANIENT outperformed ADANIPORTS but underperformed RELIANCE. The stock's Max Drawdown of −3.27% suggests that it was moderately risky this week.

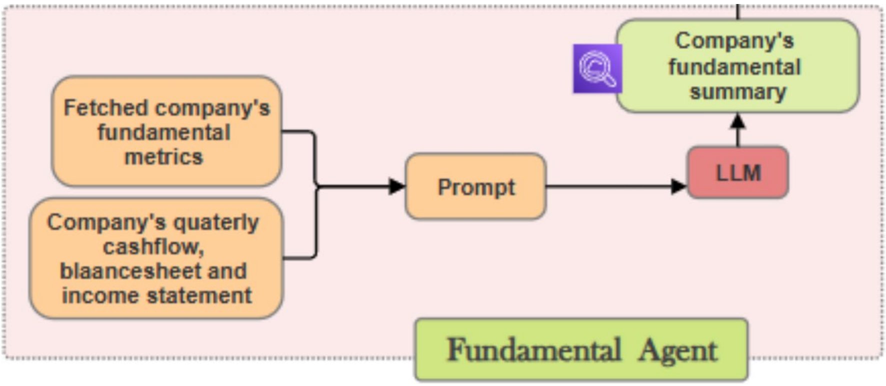

Fig. 4 Fundamental agent architecture

we assigned a sector based on NSE India data and public sources, including Yahoo Finance and Wikipedia. Companies belonging to the same industry or sub-sector were placed in the same group. For example, HDFC Bank, ICICI Bank, and Kotak Mahindra Bank are in banking sector, while Infosys, TCS, and Wipro are in Information Technology. To make this process easier, we have created a reference file that lists each company's Ticker Symbol, Full Name, and Sector. Using this, we identified the sector for each stock and selected all other NIFTY 50 companies within the same sector as its peers. This approach ensured that peer comparisons were limited to large-cap companies with similar market structures and regulatory environments.

3.4 Fundamental agent

The Fundamental Agent (Fig. 4) is responsible for analyzing the financial health and valuation of NIFTY 50 companies using structured financial statement data. The goal is to generate concise, LLM-based summaries of a company's fundamental performance, which are later used in the signal generation process. Fundamental data was collected using the `yfinance` Python package, which provides quarterly-updated financials, including income statements, balance sheets, and cash flow statements, for publicly listed companies. For each ticker, we downloaded the most recent data and saved it in structured JSON format. Since fundamental metrics do not change on a weekly basis, we reused the latest available monthly data to generate summaries for each week.

From the financial statements, we extracted and computed key valuation and performance metrics, including:

- *Revenue Growth Rate:* Year-over-year change in revenue.
- *Net Profit Margin:* Net income divided by total revenue.
- *Return on Equity (ROE):* Net income over shareholder equity.
- *Debt-to-Equity Ratio:* Total liabilities divided by shareholder equity.
- *Free Cash Flow:* Net cash from operations minus capital expenditures.
- *Earnings Per Share (EPS):* Net income divided by total outstanding shares.
- *Price-to-Earnings (P/E) Ratio:* Market price per share divided by EPS.

These indicators measure a company's financial stability and growth potential. Recent studies have demonstrated that a company's financial fundamentals have a significant impact on stock selection and allocation decisions in mutual funds. Each company's metrics were converted into a natural language prompt that presented performance across profitability, growth, leverage, cash flow, and valuation in a clear and concise format. This prompt was passed to the Llama3-70B model using the GROQ API. The model responded with a summary (example in Table 3), explaining whether the company's fundamentals were strong, weak, or showing improvement.

These summaries provide qualitative financial information that aligns with short-term price and news trends. The Signal Agent, in further processing, combines these summaries with news, price, and macro data to form the final trading signals, providing justification for each signal.

3.5 Macroeconomic agent

The Macroeconomic Agent (Fig. 5) plays a crucial role in providing a broader economic context that influences market sentiment and investor behavior when buying or selling stocks. Unlike company-specific agents, this module focuses on economy-wide metrics that can affect all sectors together, such as inflation, GDP growth, and capital flows.

We extracted high-quality macroeconomic indicators from official sources, including:

Table 3 Example of company's fundamentals summary

Company's fundamentals summary for Adani Enterprises Ltd.
Adani Enterprises Limited's financial health appears mixed. The company's profitability metrics, such as ROE and profit margins, are relatively low, indicating weaknesses in its ability to generate earnings. However, its growth metrics, including earnings growth and book value, suggest some strengths. The company's valuation, with a trailing P/E of 41.37, appears to be relatively high. Additionally, the debt-to-equity ratio of 162.59 is a significant concern, indicating a high level of leverage. Overall, the company's financial health is a mixed bag, with some positive growth trends offset by concerns regarding profitability and debt.

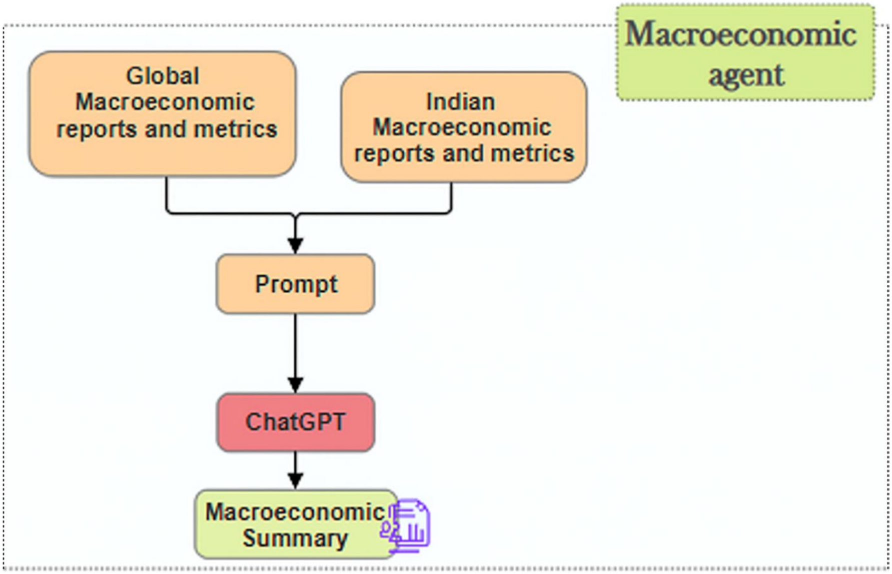

Fig. 5 Microeconomic agent architecture

- *Reserve Bank of India (RBI)*[13] – For foreign exchange reserves, reference exchange rates, inflation trends, FDI inflows, and annual macroeconomic reports.
- *World Bank*[14] – For global comparative metrics such as unemployment rate, inflation rate, and GDP growth in India, source.
- *National Stock Exchange (NSE)*[15] – For daily Foreign Institutional Investor (FII) and Domestic Institutional Investor (DII) net flows.

These sources were chosen for their reliability, frequency of updates, relevance to market participants, and trustworthiness. We focused on both domestic macroeconomic health and changes in capital flows. The main metrics considered include:

- *GDP Growth Rate:* Indicates overall economic expansion or contraction.
- *Inflation (CPI/WPI):* Rising inflation suggests increased cost pressures, often leading to interest rate hikes.
- *Unemployment Rate:* A proxy for labor market health and consumer spending ability.
- *FII/DII Net Investment:* Reflects institutional sentiment and fund inflows/outflows.
- *Foreign Exchange Reserves:* Signals external sector stability and ability to defend the rupee.
- *Currency Exchange Rates:* USD/INR fluctuations affect exporters, importers, and

[13] https://www.rbi.org.in/

[14] https://data.worldbank.org/indicator

[15] https://www.nseindia.com/reports/fii-dii

foreign capital.

- *FDI Inflows:* Tracks long-term confidence in the Indian economy by foreign businesses.

Each of these indicators has historical significance in shaping equity market changes, particularly during periods of policy changes, budget announcements, or global crises. Some macro indicators, such as FII/DII flows and exchange rates, were collected automatically via scraping and public APIs. A portion of relevant macroeconomic information is stored within unstructured documents, including the RBI's Annual Reports and Bulletins. These documents are published in PDF format and contain discussions around monetary policy, inflation management, GDP outlook, and global risks.

Due to the unavailability of structured APIs for such data and the difficulty of extracting text from scanned PDFs, we manually reviewed and summarized key insights from these reports using OpenAI's ChatGPT interface. This included: RBI's Annual Report on Macroeconomic Indicators, Inflation and Monetary Policy Outlooks, FDI and Balance of Payments Commentary.

The summaries (example in Table 4) were validated by cross-referencing official RBI charts and executive summaries. This allowed us to generate weekly macro summaries that complement both structured data (flows, forex) and policy insights. These summaries, created semi-manually, were critical in providing the LLM with an up-to-date understanding of the Indian economic scenario.

3.6 Signal agent

The Signal Agent (As shown in Fig. 1) acts as the final reasoning module in the AlphaNiftyAI pipeline, responsible for converting structured insights from multiple modalities, price, fundamentals, news sentiment, and macroeconomic conditions into a trading signal. For each NIFTY 50 company on a weekly basis, it generates a BUY, SELL, or HOLD recommendation along with a confidence score (ranging from 1 to 10), using a chain-of-thought reasoning approach via a large language model (LLM) as shown in Table 5.

To construct the input prompt for the LLM, we aggregate summaries produced by the four core agents. These include:

- A price trend summary capturing weekly return, volatility, peer-relative performance, and drawdowns.
- A financial fundamentals summary representing the most recent financial ratios such as revenue growth, ROE, EPS, and leverage.
- A news sentiment summary combining current news with past similar news events and associated price impacts.
- A macroeconomic summary highlighting key economic conditions such as FII flows, inflation, currency movement, and policy changes.

These summaries are used from previously saved paths within the system's directory. Once the full structured prompt is made, it is passed to the Llama3.3-70B model via

Table 4 Example of microeconomic summary

Microeconomic summary
Microeconomic analysis: Q3 FY23-24 (October–December 2023)
Agriculture and rural consumption:
A strong monsoon boosted foodgrain production (estimated at 3,157.7 lakh tonnes for FY23-24). MGNREGA demand dropped 0.2% y-o-y, signaling income stability. CPI inflation eased to 5.5%, with food at 6.6%, aiding rural spending.
Services Sector:
Grew 7.2%, with services exports up 21.3% to $192 billion. PMI at 58.7 reflected firm growth. GDP deflator likely moderated to 2–3%, boosting profitability.
Manufacturing & MSMEs:
IIP growth slowed to 6.1%, with manufacturing at 4.8%. WPI inflation 1–2% eased MSME costs. PLI drove $15 billion in investments.
Employment:
Unemployment at 4.2%. EPFO added 10.8 lakh jobs monthly. LFPR rose to 37% (females).
Consumer behavior:
Private consumption at 60.3% of GDP. Retail sales grew 6% y-o-y. Household expenditure shifted, with rural up 5
Automotive industry:
Two-wheeler production up 19.0%, sales up 22.6% y-o-y, reflecting demand.
Interest rates: Repo rate stable at 6.5%. Trade deficit narrowed to $70.5 billion.
Global Analysis
Global GDP: Grew 3.5% (IMF, 2023), slowing to 3.3% in 2024, with EMDEs at 4.7% vs. 4.3%.
Inflation: Global CPI eased to 6.6% (2023) from 8.7%, with U.S. at 3.4% and Eurozone at 2.9%.
Trade: World trade volume up 1.0% (2023), recovering to 3.8% (2024). Oil prices fell 10% y-o-y.
FDI: Global FDI inflows dropped 2% to $1.37 trillion (UNCTAD, 2023), with India's share at 2.6%.

the GROQ API. The prompt format is designed using the chain of thought technique, where reasoning involves asking the LLM to first interpret each section independently and then synthesize these insights to arrive at a trading decision. Specifically, the LLM is instructed to:

1. Evaluate if the price momentum indicates upward or downward strength.
2. Determine if the company's financials are in favour of growth or show weakness.
3. Analyze whether the news is positive, neutral, or negative in tone and impact.
4. Analyze if the macroeconomic environment is favorable or hostile to equity investment.
5. Combine all these insights step-by-step and provide a final signal (BUY/SELL/HOLD), with a short explanation so that the decision can be justified.
6. Provide a confidence score from 1 (low) to 10 (high), reflecting the certainty of the model's decision.

Table 5 Signal with confidence score and reason for HDFC Life

Signal	Confidence	Reason
BUY	8	The stock has a strong weekly performance with a return of 2.36%, outperforming its peers. The Sharpe Ratio of 0.86 suggests that the return was achieved with relatively low risk. The company's financial health is stable, and the news sentiment is neutral to positive, with the Max Life news being a notable positive event. Although concerns exist regarding revenue growth and debt position, the overall outlook remains positive, making it an attractive option for investors

Unlike traditional machine learning pipelines that utilize linear models or ensemble trees with feature weights, the LLM-based approach enables the Signal Agent to reason over qualitative information, such as news summaries and macro commentary, while still incorporating quantitative indicators. The chain-of-thought design enables the model to mimic the reasoning steps of a professional analyst, weighing conflicting signals, identifying macro risks, and synthesizing all of this into a single recommendation with sound reasoning. This design enables AlphaNiftyAI to produce more robust and adaptive trading decisions, particularly during high-volatility events, earnings seasons, or policy transition situations, where rigid statistical models often underperform.

4 Results and discussion

This section evaluates the performance of all strategies implemented within the AlphaNiftyAI system. We backtested each strategy using data (Refer Sect.: 3.1) from March 2024 to May 2025 (15 months) and from June 2023 to May 2025 (2 Years). We assess each strategy using standard financial evaluation metrics, with a focus on return, risk-adjusted performance, and robustness.

4.1 Evaluation metrics

The following metrics are computed to benchmark and compare strategy performance:

- *Total Return (%)*: Measures cumulative return over the evaluation period.

$$R_{\text{total}} = \left(\frac{P_{\text{end}} - P_{\text{start}}}{P_{\text{start}}} \right) \times 100 \tag{6}$$

where P_{start} is the initial portfolio value and P_{end} is the final value.

- *Sharpe Ratio*: Indicates risk-adjusted return using total volatility (Sharpe, 1966).

$$\text{Sharpe} = \frac{E[R_p - R_f]}{\sigma_p} \tag{7}$$

where R_p is the portfolio return, R_f is the risk-free rate (assumed to be 6& yearly), and σ_p is the standard deviation of portfolio returns.

- *Sortino Ratio*: Similar to Sharpe but penalizes only downside volatility (Sortino & Price, 1994).

$$\text{Sortino} = \frac{E[R_p - R_f]}{\sigma_d} \tag{8}$$

where σ_d is the standard deviation of negative returns only (downside deviation).

- *Volatility (%)*:

$$\text{Volatility} = \sigma_p \times 100 \tag{9}$$

Higher volatility indicates higher risk.

- *Win Rate (%)*: Measures how often the strategy's weekly return is higher than the benchmark's weekly return.

$$\text{WinRate} = \frac{\text{Number of periods with } R_p > R_b}{\text{Total periods}} \times 100 \tag{10}$$

where R_b is the benchmark return.

- *Maximum Drawdown (MaxDd)*: Largest peak-to-trough decline in portfolio value.

$$\text{MaxDd} = \min\left(\frac{V_t - \max_{i \le t} V_i}{\max_{i \le t} V_i}\right) \tag{11}$$

where V_t is the portfolio value at time t.

4.2 Description of strategies

- *MS*: Equally weighted portfolio rebalanced monthly/weekly based on both "buy" and "sell". signals.
- *MS-L*: Equally weighted portfolio rebalanced monthly based on the "buy" signals.
- *MS-L-Cap*: Equally weighted portfolio rebalanced monthly based on capitalization, sorted(descending), top 10 stocks based on the "buy" signals.
- *MS-Top10-LLM*: Equally weighted portfolio rebalanced monthly based on the N

stocks with the best score produced by Llama3.3-70B after processing of all the stocks with a “buy” signal.

- *MS-High-LLM*: Executes only the signals where LLM confidence score is above a threshold, 7/10, for “buy” signals, portfolio rebalanced monthly/weekly.
- *MS-TopN-cap-LLM*: Equally weighted portfolio rebalanced monthly based on the N stocks with the best score produced by Llama3.3-70B and Top N market capitalization after processing of all the stocks with a “buy” signal.
- *NIFTY++*: Equally weighted portfolio rebalanced weekly based on the “buy” signal. Applies a trailing stop-loss of 5 percent.
- *MS-TopNcap-LLM-NIFTY++*: Equally weighted portfolio rebalanced weekly based on the whole signal of stocks, where “buy” is determined by the signal and confidence is greater than 7, and from those stocks, the Top N stocks are selected with the highest market capitalization. On top of that trailing stop loss of 5 percent is integrated. The baseline method (Sect. 4.3) generates signals on a monthly basis, whereas AlphaNiftyAI produces signals on a weekly basis. Both strategies are inspired by Fatouros et al. (2024), with certain modifications incorporated.

4.3 Baseline method

We used MarketSenseAI (Fatouros et al., 2024) as a baseline method that employs a multi-agent framework that integrates diverse financial data sources, stock prices, news sentiment, company fundamentals, and macroeconomic indicators into a unified decision-making process. Data-specific agents summarize each data type, and a Signal Agent uses a large language model with chain-of-thought reasoning to produce BUY/SELL/HOLD signals and confidence scores for a particular stock.

We used the Llama 3.3-70B model instead of GPT-3 in our baseline implementation. Results are illustrated in Table 6 :Monthly.

4.4 Results of AlphaNiftyAI

The performance evaluation of AlphaNiftyAI shows its strong potential in generating risk-adjusted returns compared to the benchmark NIFTY 50 and different well-performing NIFTY 50 mutual funds. Our experiments demonstrate that incorporating weekly signals, historical news impact, and LLM-driven confidence scores significantly enhances strategy performance over time, while also reducing risk.

Table 6 Results-baseline methodology(March 2024 to May 2025)

Strategy	Return (%)	Sharpe	Sortino	Vol (%)	WinRate (%)	MaxDd (%)
MS	−1.27	−1.11	−1.52	**6.65**	46.67	**7.28**
MS-L	12.61	0.52	0.59	12.61	**66.67**	12.85
MS-L-Cap	12.71	0.55	0.65	12.71	46.67	11.76
MS-Top10-LLM	**23.57**	**1.28**	**1.52**	13.69	53.33	10.32
MS-High-LLM	12.61	0.52	0.59	12.61	**66.67**	12.85
MS-Top10-cap-LLM	12.71	0.55	0.65	12.09	46.67	11.76
NIFTY 50 Index Fund	9.17	0.24	0.28	13.4	–	15.35

Best-performing results are highlighted in bold

While the benchmark showed steady growth, strategies such as MS-Top5/10-Cap-GPT-NIFTY++ and NIFTY++ delivered higher Sharpe ratios and lower drawdowns, highlighting the benefits of capital protection through volatility-aware allocation and stop-loss mechanisms. The best return achieved by MarketSenseAI is 23.57%, whereas AlphaNiftyAI delivers a significantly higher return of 69.23%. Overall, the strategy substantially outperforms the MarketSenseAI baseline.

To align the backtesting results with real-world equity trading conditions in India, a transaction cost of 0.3% on the selling price of each stock was incorporated into the analysis. This adjustment ensures that the reported performance metrics more accurately reflect practical trading scenarios, including brokerage charges, taxes, and other execution-related costs typically incurred by retail and institutional investors. Results are illustrated in Tables 8 and 10.

Overall, the results validate that AlphaNiftyAI provides a more granular, adaptive, and resilient approach to investment decision-making compared to traditional monthly signal-based methods. Illustrated in Tables 7 and 9.

4.5 Comparative analysis

Key insights drawn from the Tables 6, 7 ,9 and 11 are given below. The shift to weekly signals captured more granular trends and market movements, and returns from each strategy became more stable and reliable.

The proposed strategy MS-Top5-cap-LLM-NIFTY++ achieved the highest total return, Sharpe, and Sortino, with the lowest max drawdown, clearly outperforming all other approaches in both the time periods as in Tables 7 and 9.

NIFTY++ alone showed significant improvement in both risk and return, validating the utility of stop-loss. Strategies with stop-losses have a lower maximum drawdown value, which means the risk of loss is reduced.

Compared to leading NIFTY 50 index funds in India (Table 11), most strategies yield superior returns, exhibit better drawdowns, and have risk-adjusted profiles. It can be observed in the graphs (Figs. 6, 7, 8, 9) that the strategies which has stop loss are performing much better than the some strategies, particularly the simpler LLM-based ones without volatility filters, underperformed because they tend to include high-volatility stocks that amplify drawdowns even when the prediction signal is

Table 7 Results-AlphaNiftyAI (data from March 2024 to May 2025)

Strategy	Return (%)	Sharpe	Sortino	Vol (%)	WinRate (%)	MaxDd (%)
MS	9.76	0.28	0.52	**13.66**	50.77	8.85
MS-L	10.20	0.25	0.41	16.75	42.42	11.37
MS-L-Cap	13.20	0.44	0.73	16.60	44.62	10.20
MS-Top10-LLM	9.63	0.22	0.35	16.89	42.42	11.97
MS-High-LLM	10.20	0.25	0.41	16.75	42.22	11.37
MS-Top5-cap-LLM	19.33	0.80	0.64	15.49	48.48	10.29
NIFTY++	58.62	3.06	6.91	17.19	56.06	5.47
MS-Top10-cap-LLM-NIFTY++	62.97	3.22	7.20	17.08	59.09	5.23
MS-Top5-Cap-LLM-NIFTY++	**69.23**	**3.71**	**8.62**	17.06	**62.12**	**4.56**

Best-performing results are highlighted in bold

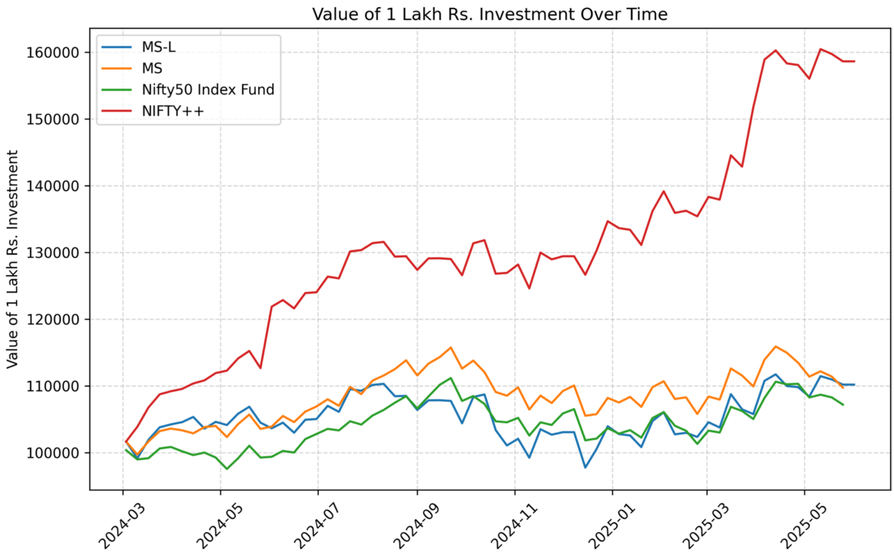

Fig. 6 Performance of only LLM signal strategies VS NIFTY 50 index fund

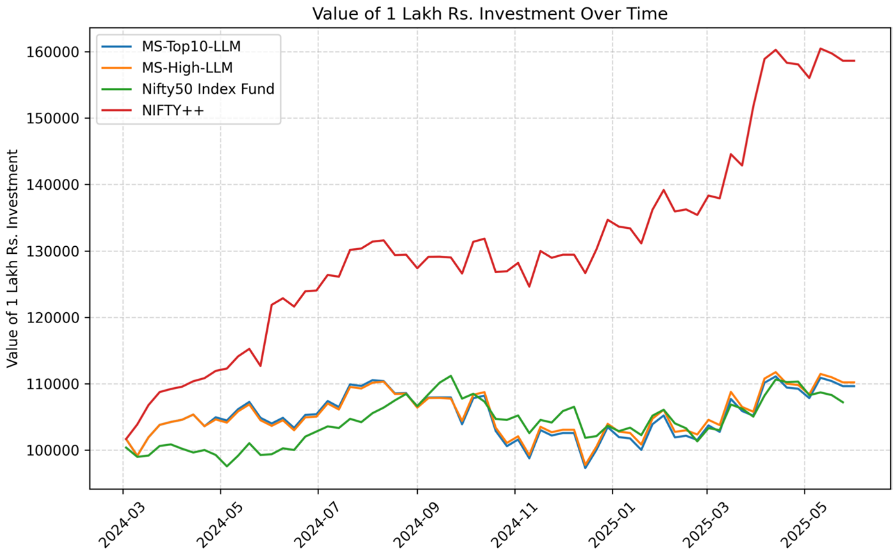

Fig. 7 Performance of only LLM signal +confidence strategies VS NIFTY 50 index fund

correct. Strategies that relied solely on BUY/HOLD decisions from LLM summaries (without macro-awareness or volatility features) struggled during sudden market reversals, indicating that sentiment alone is insufficient for robust weekly trading.

Approaches such as MS, MS-L, and MS-L-Cap underperformed due to a lack of risk adaptation. These strategies often held stocks through periods of elevated volatility, resulting in weaker risk-adjusted returns. Extending the evaluation time period

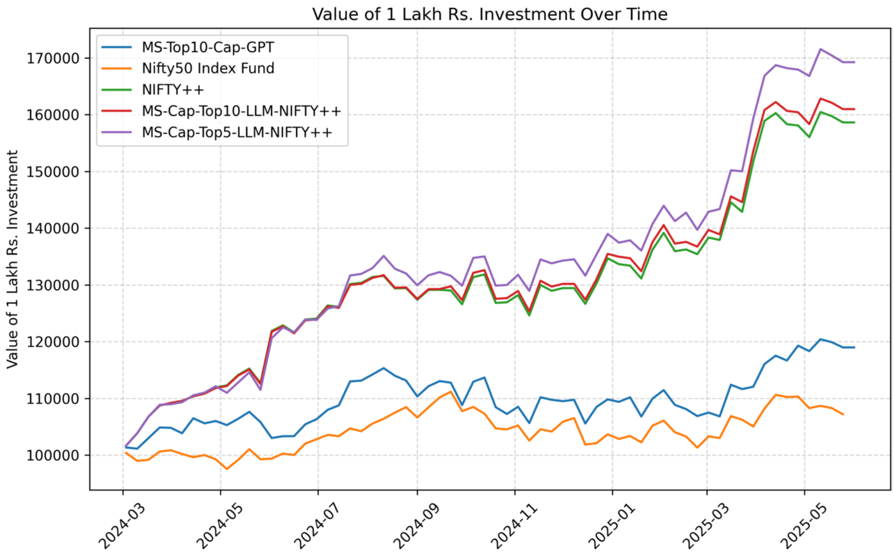

Fig. 8 Performance of LLM signal+confidence+cap+stoploss strategies VS NIFTY 50 index fund

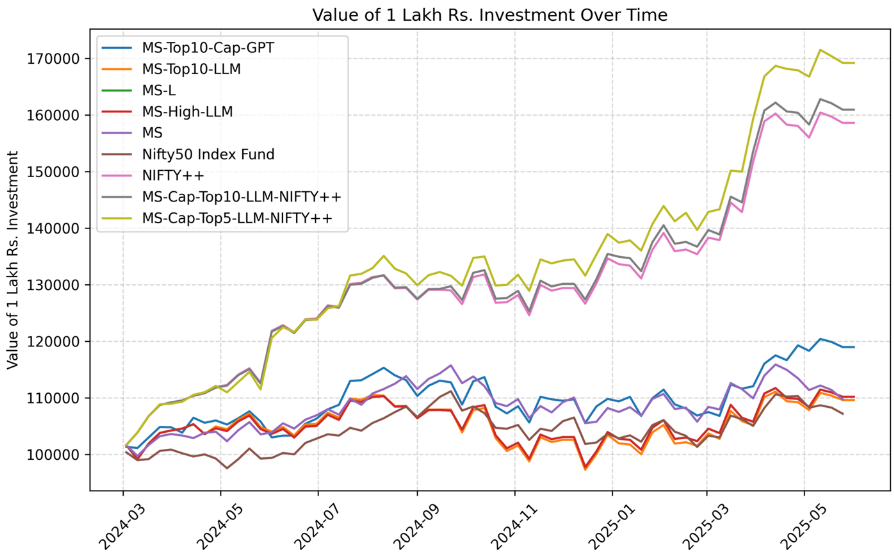

Fig. 9 Performance of all strategies together

does not affect the strategy's performance, as the results remain consistent, which strengthens confidence in the framework. The strategies with stop-loss have very low maximum drawdown, which indicates that the risk of loss is lower in them.

Models without confidence-based filtering admitted lower-quality LLM signals into the portfolio, which reduced the precision of stock selection and contributed to

Table 8 Results-AlphaNiftyAI (data from March 2024 to May 2025) with transaction cost of 0.3% on selling price of each stock

Strategy	Return (%)	Sharpe	Sortino	Vol (%)	WinRate (%)	MaxDd (%)
MS	−10.58	−1.21	−2.05	**13.66**	50.77	4.66
MS-L	−10.03	−0.96	−1.53	16.75	42.42	16.48
MS-L-Cap	−8.02	−0.84	−1.35	16.62	44.62	15.87
MS-Top10-LLM	−10.50	−1.18	−1.98	16.00	42.42	17.12
MS-High-LLM	−10.03	−0.96	−1.53	16.75	42.22	16.48
MS-Top5-cap-LLM	−0.76	−0.40	−0.64	16.87	50.00	15.03
NIFTY++	29.64	1.38	2.94	17.19	56.06	9.07
MS-Top10-cap-LLM-NIFTY++	30.82	1.45	3.06	17.08	56.06	8.72
MS-Top5-Cap-LLM-NIFTY++	**40.13**	**1.97**	**4.17**	17.33	**62.12**	8.35

Best-performing results are highlighted in bold

Table 9 Results-AlphaNiftyAI (data from June 2023 to May 2025)

Strategy	Return (%)	Sharpe	Sortino	Vol (%)	WinRate (%)	MaxDd (%)
MS	31.93	1.97	3.36	**13.13**	53.85	8.83
MS-L	43.57	2.25	3.74	16.67	49.52	11.37
MS-L-Cap	46.76	2.46	4.10	16.60	50.96	10.81
MS-Top10-LLM	42.83	2.20	3.61	16.76	49.52	11.97
MS-High-LLM	43.57	2.25	3.74	16.67	49.52	11.37
MS-Top5-cap-LLM	58.32	3.13	5.18	16.71	51.43	10.27
NIFTY++	110.78	6.27	12.52	16.72	58.10	6.49
MS-Top10-cap-LLM-NIFTY++	112.71	6.40	12.71	16.68	55.24	6.49
MS-Top5-Cap-LLM-NIFTY++	**127.80**	**7.24**	**14.42**	16.83	**60.00**	**6.49**

Best-performing results are highlighted in bold

weaker performance. To replicate the real-time situation, we have added transaction cost of 0.3% for selling each stock.

Overall, the results indicate that the combination of (1) LLM-generated signals, (2) Confidence score, and (3) stop-loss rules leads to a more robust and superior investment framework compared to both baseline models and industry benchmark index funds.

4.6 Comparative analysis after transaction cost

Key insights drawn from the Tables 6, 8 , 10 and 11 are as follows : Introducing a transaction cost of 0.3% makes the backtest more realistic for Indian equity markets, where brokerage, STT, exchange fees, and slippage together add up to this level. After including costs, the overall returns of MS-Top5-Cap-LLM-NIFTY++ decrease slightly, but the strategy still performs well, demonstrating that its strong signal generation helps offset trading expenses. Strategies with higher turnover (e.g., those without stop-loss or without filtering by confidence) experience a larger drop in net returns, suggesting that frequent trading reduces profitability when realistic costs are taken into account.

The NIFTY++ variants demonstrate greater resilience to transaction charges, as their stop-loss feature mitigates over-trading by automatically cutting positions

Table 10 Results-AlphaNiftyAI (data from June 2023 to May 2025) with transaction cost of 0.3% on selling price of each stock

Strategy	Return (%)	Sharpe	Sortino	Vol (%)	WinRate (%)	MaxDd (%)
MS	−3.4	−1.17	−1.87	**13.13**	53.85	14.66
MS-L	7.38	−0.28	−0.45	16.75	49.52	16.48
MS-L-Cap	9.78	−0.13	−0.22	16.6	50.96	15.87
MS-Top10-LLM	6.83	0.04	0.07	16.89	42.42	17.12
MS-High-LLM	7.38	−0.28	−0.45	16.75	49.52	16.48
MS-Top5-cap-LLM	18.45	0.39	0.63	16.69	51.43	15.03
NIFTY++	57.84	2.75	5.35	16.64	58.1	9.07
MS-Top10-cap-LLM-NIFTY++	59.28	2.84	5.48	16.62	55.24	8.72
MS-Top5-Cap-LLM-NIFTY++	**70.62**	**3.49**	**6.74**	**16.77**	**60**	**8.35**

Best-performing results are highlighted in bold

Table 11 Nifty 50 index funds comparison (June 2023 to May 2025)

Fund	Total return (%)	Sharpe	Sortino	Vol (%)	WinRate (%)	MaxDd (%)
ICICI Nifty50	40.91	0.90	1.20	13.04	54.79	15.51
Nippon Nifty50	55.95	1.17	1.38	14.55	60.67	19.54
SBI Nifty50	60.24	1.73	2.30	10.29	60.78	8.46
Kotak Nifty50	60.24	1.73	2.30	10.29	60.78	8.46
Axis Nifty50	45.74	1.04	1.40	12.85	57.00	16.48

early and preventing unnecessary trades. Compared to NIFTY 50 index funds, even after adding a 0.3% cost, the AI-driven strategies still provide competitive or better returns, demonstrating their ability to perform well in changing market conditions. The similarity in net returns to leading mutual funds indicates that AI-based selection, although rule-based and systematic, can match those of professionally managed funds while maintaining transparent decision-making.

5 Avoiding overfitting and ensuring model robustness

A key challenge in any data-driven financial strategy is the risk of overfitting, where a model performs extremely well on historical data but fails to generalize to new, unseen market conditions. While developing our AlphaNiftyAI framework, several safeguards were implemented to mitigate this risk and enhance the model's real-world robustness. These practices also align with the principles discussed in the "Seven Sins of Quantitative Investing" (Hsu & Kalesnik, 2014).

First, the model relies on weekly data spanning an extended period, covering diverse market conditions, macroeconomic cycles, and varying volatility conditions. This helps ensure that the signals are not made for a narrow or unusually favorable time window. Second, all components of the system, which are price trends, fundamentals, news, and macroeconomic factors, are created separately before being combined into a signal. This design prevents the algorithm from learning spurious correlations or overly complex relationships that may not persist over time.

Third, the LLM outputs are confined to a uniform structure and limited to important qualitative reasoning rather than numerical optimization. Since the model does not tune parameters to maximize backtest performance, it reduces the likelihood of overfitting. Fourth, the system was initially evaluated on data from March 2024 to May 2025, a second, longer backtest was conducted from June 2023 to May 2025. Consistent results across different time spans demonstrate that the system's behaviour is stable and not limited to a specific window of market conditions.

Finally, the framework shows transparency and interpretability. Each decision is accompanied by a natural language explanation justifying the decision, enabling researchers and investors to identify unrealistic assumptions or patterns that may indicate overfitting. These measures collectively help ensure that AlphaNiftyAI avoids the common pitfalls highlighted in the literature, including data mining, excessive complexity, and unstable relationships, thereby improving its reliability in real-world use.

6 Conclusion

In this work, we present AlphaNiftyAI, a fully automated, end-to-end investment analysis and signal generation system that leverages the power of large language models (LLMs), structured financial data, macroeconomic indicators, and historical price-impact memory. By systematically integrating multiple data modalities, including stock prices, company fundamentals, financial news, and macroeconomic context, our framework generates explainable BUY/SELL/HOLD signals with associated confidence scores. We also implement both monthly and weekly pipelines, backtest various strategies, including MS-L, NIFTY++, and LLM-confidence-prioritized portfolios, and compare their performance against benchmarks such as the NSEI. Our results show that LLM-based strategies not only match but sometimes outperform traditional benchmarks and even expert-handled mutual funds, particularly when enhanced with volatility filters, sentiment-driven news summaries, and macroeconomic involvement. By utilizing similar news memory and trailing stop-loss logic, AlphaNiftyAI bridges the gap between research and real-world trading scenarios. Overall, this work contributes a scalable and transparent approach to LLM-driven investing, offering a strong foundation for future research and applications in AI-augmented financial decision-making. The enhancements made to the original LLM-driven trading approach, including the introduction of higher-frequency (weekly) signals, confidence-based filtering, peer relative analysis, and practical constraints such as stop-loss, have produced a more effective and realistic trading system. AlphaNiftyAI demonstrates that with structured prompting, signal interpretation, and proper post-processing logic, LLMs can be integrated reliably in financial forecasting systems.

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Data availability The data that support the findings of thisstudy are available from the first author upon reasonable request.

Declarations

Conflict of interest The authors declare no conflict of interest.

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